

# Haocheng He

Phone: 56987112 | haocheng.he@foxmail.com | Hong Kong SAR, China  
linkedin.com/in/haocheng-he-92657028b

## PROFESSIONAL SUMMARY

Industrial Engineering graduate from Guangdong University of Technology, with a minor bachelor's degree in Law, and currently pursuing an MSc in Aviation Engineering and Operations Management at The Hong Kong Polytechnic University. My experience spans four internships in AI hardware integration, airport emergency simulation, manufacturing improvement, and assembly-line optimization, along with a time-series RUL prediction project. I learn quickly, enjoy digging into technical problems, and am comfortable working across hands-on engineering, data modeling, and high-pressure execution.

## EDUCATION

**The Hong Kong Polytechnic University** Hong Kong SAR, China  
*Master of Science in Aviation Engineering and Operation Management* Sep 2025 – Jan 2027  
GPA: 3.95 / 4.3 (Current)  
Thesis: Simulation Modelling and Optimization for Airport Ferry Vehicle Scheduling under Uncertainty

**Guangdong University of Technology** Guangzhou, China  
*Bachelor of Engineering in Industrial Engineering* Sep 2021 – Jun 2025  
GPA: 3.67 (Top 20%)  
Thesis: Optimization simulation of Route Planning for emergency special Vehicles in large Airports Based on Anylogic  
Honors: Excellent Graduation Thesis + Innovation Award

*Bachelor of Laws (Minor Degree in Law)* Sep 2022 – Jun 2025  
GPA: 3.61  
Thesis: Coordination Between Copyright Protection and Privacy Protection in Digital Rights Management

## EXPERIENCE

**AI Engineer Assistant** May 2025 – Jun 2026  
*Shenzhen Wuxian Hongyin Technology Co., Ltd.* Shenzhen, China

- Participated in smart piano R&D based on Huawei Ascend 310B, covering PCB hardware tests (CPLD/STM32), piano motor tests, full-bridge motor driver schematic/layout, and STM32 firmware development.
- Supported partial module integration for a camera- and Ascend-based piano fingering recognition system; reviewed code, understood system architecture, and helped locate issues.
- Maintained Ubuntu servers and Huawei switches/routers; also supported procurement, administration, and technical interviews.

**Research Assistant** Sep 2023 – Jun 2025  
*Guangdong University of Technology (with Guangzhou Baiyun Int'l Airport)* Guangzhou, China

- Built an airport road-network, apron, and runway-area simulation model in AnyLogic, Java, and MySQL for aircraft landing and emergency fire-truck dispatch scenarios.
- Designed single/multi-vehicle, single/multi-event, and runway-road-closure scenarios; compared D\* Lite, A\*, Dijkstra, BFS, and GA routing algorithms.
- Completed 4 emergency-scenario comparisons across 5 routing algorithms; D\* Lite achieved the highest average reliability index (1.342).

**Industrial Engineering Intern** Apr 2024 – May 2024  
*Midea Group — Kitchen & Water Heater Division* Foshan, China

- Conducted stopwatch time studies across all workstations on a dishwasher sheet-metal line, built operator workload tables, and identified 5 over-cycle steps.
- Proposed 22 improvements using lean methods; simulation showed line balance improving from 58% to 70%.

reducing headcount from 20 to 18.

**Manufacturing Operations Intern**  
*Guangxi LiuGong Machinery Co., Ltd.*

Jul 2023 – Aug 2023  
*Liuzhou, China*

- Timed 12 workstations on the wheel loader assembly line with a stopwatch; identified 3 major bottlenecks using ECRS analysis.
- Proposed layout adjustments estimated to reduce cycle time by approximately 8% in simulation. Discussed findings with line engineers.

## PROJECTS

---

**Aircraft Engine Remaining Useful Life Prediction (Course Project)**

Python / C-MAPSS

- Built an RUL prediction workflow on NASA C-MAPSS FD001 multivariate time-series data; cleaned operating-condition data and 21 sensor signals, constructed sequence features, and applied PCA.
- Compared LSTM, BiLSTM, Transformer, and Random Forest baselines for time-series forecasting; after iterative tuning, the BiLSTM model reached about 8% final error from an initial model error of about 50%.

## SKILLS

---

**Programming & Data:** Python, Java, SQL/MySQL, C/STM32, JavaScript/TypeScript

**AI & Automation:** Claude Code, Codex, Hermes, ChatGPT, Prompt Engineering, Harness Loop Engineering, AI Agent workflows, model deployment/evaluation

**Machine Learning:** Random Forest, LSTM, BiLSTM, Transformer, PCA, model training/evaluation

**Simulation & Optimization:** AnyLogic, Plant Simulation, OR-Tools, Dijkstra, A\*, BFS, genetic algorithms, line balancing, ECRS

**Engineering Tools:** Ubuntu/Linux, Git, LCEDA, oscilloscope, logic analyzer, Huawei switches/routers

**Languages:** English (IELTS 6.5), Mandarin (native), Cantonese (conversational)

## PUBLICATIONS & INTELLECTUAL PROPERTY

---

### Invention Patents

- PI Coating Machine Motion Control via MCMC Sampling — Zhou Xianzhong, He Haocheng, He Zhiyuan (2026). CN122085794A, Guangdong University of Technology.
- Coating Control Method and System (Deep Reinforcement Learning) — Zhou Xianzhong, He Haocheng, He Zhiyuan (2025). CN121325552A, Guangdong University of Technology.
- Home Solar Energy Storage System — Adaptive Energy Regulation — Zhou Xianzhong, Zeng Huangtao, Yu Daming, He Haocheng (2024). CN117767437A, Guangdong University of Technology.

### Utility Model

- Integrated Wire Harness Production Equipment — Zhao Xiang, He Haocheng, Hu Bofeng. CN221352455U.

### Software Copyright

- Enterprise Financial Data Analysis Software V1.0 — Li Yisi, He Haocheng (2024). Reg. No. 2024SR0146827.
- Urban Flood Intelligent Mobile Emergency Command Dispatch System V1.0 — Lin Chengfeng, Li Zeyin, Yang Hua, He Haocheng (2024). Reg. No. 2024SR0549247.